3.2.2 Verifying problems by casting out 9

Any number or combination of digits in that number which add to 9 can be cast out from the given number. Then the sum of remaining digits of the number is divisible by 9 or multiple of 9.

Example 1

Is 46908 multiple of 9?

$46\cancel{9}08 = 4+6+0+8$

$= 18$

$= 1 + 8$

$= 9$

46908 is a multiple of 9 or divisible by nine.

Example 2

Check the following numbers whether it is a multiple of 9 or not

$24689 = 2 + 4 + 6 + 8 = 20$ (It is not a multiple of 9.)

$\cancel{9}1\cancel{0}8 = 0$ (It is a multiple of 9.)

$3\cancel{1}\cancel{6}5 = 1 + 5 = 6$ (It is not a multiple of 9.)

Addition

In addition problem, we can check the sum by casting out nines.

Example 3

Check the addition fact

$3356 + 4729 = 8085$

$3\cancel{3}5\cancel{6} + 4\cancel{7}2\cancel{9} = 8085$

$8 + 4 = 21$

$12 = 21$

$1 + 2 = 2 + 1$

$3 = 3$